Online Activism and Dyadic Representation: Evidence from the UK E-Petition System

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Abstract

By making it easier for citizens to communicate their preferences, online forms of political participation have the potential to strengthen the representational link between politicians and voters. However, we know little about the effects of online advocacy on politicians' behaviour. Using new data from an e-petition system in the UK, I show that support for a petition amongst a Member of Parliament's constituents substantially increases the probability that the MP advocates for the petition in parliamentary debate, even when compared to their behaviour in counterfactual non-petition debates which focus on the same policy issues. However, MP responsiveness is conditioned both by party discipline and electoral competition. These findings have important implications for our understanding of dyadic representation in parliamentary systems.

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Normative theories of democratic representation suggest that, when viewed as delegates, politicians should advance the preferences of their constituents (Pitkin 1967). In single-member district (SMD) electoral systems, a key concern is whether legislators' actions reflect the specific views of their districts – a concept usually described as "dyadic" representation (Weissberg 1978). However, a central dilemma for reelection-seeking politicians is that acquiring information on the preferences of voters can be difficult and costly. This can have pernicious effects, for even when politicians have incentives to adapt their behaviour to reflect constituents' preferences, a paucity of information will mean that politicians' knowledge of such preferences may be biased (Kingdon 1967, Broockman & Skovron 2018); responsiveness will be limited (Stimson et al. 1995); and that the quality of democratic representation is likely to suffer (Mansbridge 2003, 518). Put bluntly, it is hard to do what the voters want if you do not know what it is that the voters want you to do.

The growth of online political activity therefore represents an important change to the information environment faced by politicians. By lowering the costs of acquiring political knowledge and giving voters new mechanisms for communicating with their representatives, the internet has the potential to improve the informational link between politicians and voters. For some scholars, this promises to strengthen representation and "enrich democracy." (Coleman & Blumler 2009, 11) Likewise, politicians have also argued that the internet can be used to "make government more open and responsive than ever before [by] empowering citizens with new ways to participate in their democracy" (Obama 2011). However, despite these claims, no empirical work to date has evaluated whether new online forms of political expression have had tangible effects on the quality of democratic representation. This paper helps to fill this gap.

In particular, I investigate the effects on dyadic representation of one popular form of online democratic engagement: government-sponsored electronic petition (e-petition) systems. I argue that e-petitions represent a valuable source of information on constituent preferences for reelection-seeking politicians. Legislators typically learn about constituent preferences through imperfect signals – emails, letters, social media, etc – which makes it difficult to form correct judgements of constituency opinion, and potentially inhibits legislator responsiveness.

The petition data I study, by contrast, provides legislators with a natural metric for measuring the strength of district opinion across a variety of issues. As petition signature counts are commonly available at the electoral district level, legislators can use this information to guide their actions in ways that are likely to be electorally rewarding, and in so doing may improve the representation of constituent interests in the policy process.

The paper makes contributions both to the study of e-petitions, and to the literature on dyadic representation in parliamentary systems. First, the UK e-petition system that I study has proved exceptionally popular. E-petitions have collectively accumulated nearly 50 million signatures since 2015, and petition-signing is the second most popular form of political activity (after voting) in the UK, with nearly 30% of adults report signing an e-petition in the past year (Hansard Society 2018). While both academics and policymakers see e-petitions as a method for strengthening the representational bond between representatives and their constituents (Hough 2012, Bochel 2012, Procedure Committee 2014), the empirical literature has provided only descriptive evidence of the petitioning process (Kennedy Stewart et al. 2013, Jungherr & Jürgens 2010, Lindner & Riehm 2011, Escher & Riehm 2017, Yasseri et al. 2017, Hersh & Schaffner 2018). Given the rapid adoption of these systems in many countries in recent years,[1] it is surprising that no study to date has addressed the representational effects of these systems. I do so here.

Second, existing studies of dyadic representation in parliamentary systems either rely on demographic proxies for measuring district opinion (Soroka et al. 2009, Blidook & Kerby 2011), or are limited to evaluating the dyadic relationship on a small number of high-profile issues that are covered in national opinion polls (Hanretty et al. 2016). No previous work has linked observational evidence of constituent activism with legislative behaviour, which is troubling because, as Stimson et al. (1995, 562) argue, "while we, as scholars, rely solely on public opinion polls to do our work, politicians surely do not." E-petitions provide legislators with clear, quantitative and district-specific information about the relative salience of a wide variety

[1] E-petition schemes exist in the UK (https://petition.parliament.uk), the US (https://petitions.whitehouse.gov), Germany (https://epetitionen.bundestag.de), Canada (https://petitions.ourcommons.ca), Australia (https://www.aph.gov.au/Parliamentary_Business/Petitions), the Ukraine (https://petitions.com.ua), as well as in the European Parliament (https://petiport.secure.europarl.europa.eu/petitions) and in many local government settings (Hough 2012).

of political issues, and – because petition signature counts are publicly available – they also offer a rare opportunity for researchers to observe the same indicators of constituency-demand as those observed by politicians.

Third, while most studies evaluate whether legislators cater to district preferences by adapting their roll-call *voting* behaviour (Miller & Stokes 1963, Erikson 1978, Kastellec et al. 2010, Krimmel et al. 2016), strong party discipline means that dyadic voting is unlikely to manifest in most parliamentary systems (Powell Jr 2004, 283-285). As a consequence, scholars have had to rely on evidence taken from non-typical roll-call votes in parliamentary systems, such as those where no whip is imposed, or in the rare instances where votes cut across party lines (Hanretty et al. 2016). The UK petition system suggests an alternative setting for dyadic representation: in parliamentary debates. A critical feature of this system is that e-petitions receiving more than 100,000 signatures are considered for debate in parliament. In addition to exhibiting more variation than legislators' voting decisions, an MP's decision to participate in plenary debate – the main dependent variable of interest here – constitutes an important representative act, as making a speech in a debate of great local interest allows an MP to literally 'speak up' for her constituents' interests.

The expectation I test in this paper is straightforward: MPs will be more likely to participate in debates on petitions marked by high levels of local support. Identifying such an effect, however, is complicated by the fact that if an MP already has a strong sense of which issues are important to her constituents, she might participate in a given petition debate even in the absence of the information provided by the relevant e-petition. An MP from an agricultural constituency, for example, will likely receive many signatures on agricultural petitions, will probably attend the relevant petition debates, but will probably also attend *non-petition* agricultural debates. In such situations, the relationship between petitions signatures and debate behaviour will be confounded. To overcome this identification concern, I describe an empirical strategy where, by estimating each MP's tendency to speak in many debates on certain topics, I am able to examine their responsiveness to e-petitions net of these more general representational behaviours. Combining statistical topic models with a multilevel random-effects model, I show that MPs' responsiveness to constituent activism in petition debates differs from their

behaviour in counterfactual non-petition debates that are otherwise very similar in terms of topical content.

I report three main results. First, I show that legislators are substantially more likely to participate in debates relating to petitions signed by many of their constituents. Second, while this effect is robust to controlling for participation on topically-similar non-petition debates, MP responsiveness is clearly conditioned by both party discipline and electoral competitiveness. In particular, MPs representing competitive constituencies are more responsive to petition signatures than are MPs in 'safe' seats, and newly elected MPs are also more responsive than more experienced MPs. On the other hand, MPs in 'frontbench' positions are less responsive to petition support than are backbench MPs. Finally, I provide evidence that MPs not only participate more in relevant debates, but are also more likely to use their speeches to *express agreement* for e-petitions marked by stronger local support. Taken together, these findings suggest that e-democracy initiatives can significantly the strengthen representational ties between citizens and politicians, but that the effects of such initiatives are also likely to be mediated by existing political incentives.

Information, Dyadic Representation, and Parliamentary Debate

Positive accounts of representation suggest that reelection-seeking politicians will tailor their behaviour to satisfy the wishes of voters, either by taking positions that align with district preferences, or by allocating attention to the most salient district-specific issues (Downs 1957, Mayhew 1974, Cain et al. 1984, Hall 1996). Information is a central component of these theories (Mansbridge 2003, 516–520), as legislators are expected to infer the preferences of their constituents, and use this knowledge to guide their legislative decisions. However, legislators will only occasionally have access to high quality information (such as district-specific opinion polls) to make inferences about the opinions of their constituents on particular policy issues. More commonly, legislators must rely on heuristics to evaluate the strength of opinion towards a specific policy in their constituencies: how much correspondence has the legislative office received? How prominent is the issue in local media? It is therefore unsurprising that politicians invest significant resources in gathering additional information about constituent

preferences (Maestas 2003), and that even small changes in this information can translate into meaningful changes in legislative behaviour (Butler et al. 2017, Bergan & Cole 2015).

For legislators, the value of online political expression, and of e-petitions in particular, resides in the additional signal it offers about constituents' preferences. First, e-petitions provide a clear quantitative measure of the relative importance of different issues. The number of signatures a petition receives can be easily compared to the signature rate on another petition, allowing legislators to make simple comparisons of salience across issues. This information can then be used to update legislators' beliefs about the best ways to allocate their time to different issue areas.

Second, e-petition signature counts are typically available at low geographic levels of aggregation. This is critical, as locally-disaggregated information allows legislators to respond to the specific preferences of their constituents, which may differ from those of the population at large. Such information is likely to be especially valuable in the UK, where constituency-specific opinion polls are rare (Wring et al. 2016). Furthermore, the search costs for acquiring this information are small: E-petitions are made publicly available on a dedicated website where anyone can sign by providing their name, email address and postcode. Petition signatures – which are publicly viewable – are are disaggregated by constituency, and for any petition that receives over 100 thousand signatures the parliamentary Petitions Committee also provides this information directly to MPs (Procedure Committee 2014, 48).

Third, e-petitions tend to focus on a more diverse set of topics than those that traditionally dominate national debate (Hersh & Schaffner 2018), and which also contrasts with the limited range of issues considered by national opinion polls. In the UK, e-petitions address a wide variety of issues – both salient and non-salient – on which it is unlikely that MPs will always have strong prior understanding of constituent preferences. For instance, the petition calling on the government to "Stop allowing immigrants into the UK"[2] is typical of the type of issue that might feature in UK election campaigns. By contrast, the petition that lobbied the government to "Give the Meningitis B vaccine to ALL children"[3] addressed an issue that is not typically central to UK political debate, and where MPs were likely to have weaker

[2] https://petition.parliament.uk/archived/petitions/106477
[3] https://petition.parliament.uk/archived/petitions/108072

intuitions about the importance of the issue to their constituents. E-petitions may therefore also be informative about constituent opinion on issues that have not previously been a part of the political agenda.

Most studies of dyadic representation focus on the congruence between constituency opinion and legislators' *voting* behaviour (Miller & Stokes 1963, Erikson 1978, Kastellec et al. 2010, Krimmel et al. 2016). However, while e-petitions may facilitate representation "by providing a mechanism to enable the public to express their views to those in elected representative institutions" (Bochel 2012, 798), their informational value in parliamentary systems may be mitigated by countervailing incentives to toe the party line. An inherent tension in SMD parliamentary systems is that "electoral rules create incentives for a personal vote and legislative rules deny representatives the opportunity to establish it." (Cain et al. 1984, 217) In particular, legislators in parliamentary systems will be less able to use roll-call votes as a tool for demonstrating responsiveness than is the case for legislators in presidential systems (Carey 2007). Therefore, the provision of constituency-specific information may generate significant incentives for dyadic behaviour, but the party-discipline constraints in parliamentary systems may mean that legislators are unable to respond to such information.

However, the UK petition system is formally integrated within the political process in Westminster and it provides MPs with an alternative mechanism for responding to petitioners' concerns. In particular, any petition that receives over 100,000 signatures nationally is considered for debate in parliament. Although the Petitions Committee has some discretion over which debates are scheduled, almost all petitions crossing the 100,000 threshold (as well as some slightly below it) have been debated.[4] These debates have become a high-profile part of the parliamentary week, and are attended by members of all political parties as well as the government minister responsible for the relevant department.

How might we examine responsiveness in the context of these debates? There is no substantive vote at the end of petition debates and rather than focussing solely on the positions that MPs take when they participate in debate, I consider the decision to participate in a given debate to be itself a representational act. Politicians may respond to new information

[4]Table S6 in the appendix lists the 84 debates held between June 2015 and February 2019.

about constituent preferences not only by changing their policy positions (as revealed by the votes they cast), but rather by shifting their legislative efforts towards issues that their constituents find most salient. As Hall (1996, 58) suggests, constituency influence should not only operate on legislators' voting behaviour "but on the intensities that they reveal in their decisions about when and to what extent they will participate in particular matters". By allocating more of their time to the most pressing district issues, politicians demonstrate their willingness to address constituent concerns.

Legislative debate is a useful arena for pursuing this type of representational strategy, as parliamentary speeches are publicly visible acts that are easy to publicise to constituents. Parliamentary debates are widely covered in the UK media, and both voters and MPs consider participating in debate to be one of the most important roles that an MP performs (Campbell & Lovenduski 2014). Speaking in a debate, then, is a credible signal to constituents of the importance an MP assigns to a given issue in parliament. Standing up and making a speech in a debate devoted to a petition which many of your constituents have signed allows the MPs to demonstrate that she has heard local concerns, and has given voice to those concerns in parliament. Additionally, speechmaking is less subject to control by party elites than the votes cast in roll-calls (Proksch & Slapin 2012). If strong party discipline in parliamentary systems does mitigate incentives for dyadic representation, the fact that petition debates are not subject to roll-call votes suggests that MPs may be able to respond to the preferences of their constituents without challenging the party line.

In sum, the UK e-petition system provides MPs with clear information regarding constituent issue priorities, and a formal mechanism for responding to that information that is not generally subject to strict party discipline. I therefore expect that MPs will be more likely to participate in debates relating to petitions marked by high levels of local support.

Hypothesis 1: *MPs are more likely to participate in relevant debates for petitions signed by many of their constituents than for petitions signed by few of their constituents.*

Beyond this central expectation, I also expect the strength of the dyadic link to vary according to electoral and party discipline incentives. For example, an MP's responsiveness

to petitioners' demands is likely to vary as a function of the electoral environment the MP faces. Politicians who win office in districts with small electoral margins have greater incentive to respond to constituents than politicians who win office by large margins (MacRae 1952), and, in the US, the dyadic connection is indeed stronger for legislators in more competitive districts (Griffin 2006). I expect the same dynamics to operate in the UK.

Hypothesis 2: *The relationship between petition signatures and participation will be stronger for MPs in more electorally competitive constituencies.*

Similarly, MPs who have recently entered parliament are also likely to be more eager to establish a dyadic connection with their constituents than their more experienced colleagues. New MPs have had less time to learn about constituent issue priorities, making the information from petitions more valuable. In addition, new MPs are significantly less time constrained, as they are less likely to hold frontbench positions or to serve on time-intensive select committees. New MPs are also less likely to have established a reputation for standing up for their constituents, and so should be more willing to put on record their support for issues that have attracted significant local support.

Hypothesis 3: *The relationship between petition signatures and participation will be stronger for MPs who have recently entered parliament.*

By contrast, MPs who hold frontbench positions for either the government or the main opposition party are more likely to be responsive to demands from party leaders than are backbench MPs (Benedetto & Hix 2007), and frontbench MPs are formally responsible for attending certain debates in the House (those pertaining to the jurisdiction of their ministries) and so have less freedom to choose which debates to attend. Accordingly, the relationship between petition signatures and participation should be weaker for frontbench MPs than for backbench MPs.

Hypothesis 4: *The relationship between petition signatures and participation will be weaker for MPs who hold frontbench positions.*

Data

I use data on 108 petitions debated between June 2015 and February 2019, which I link to debate transcripts pertaining to each petition. The Petitions Committee will occasionally group two petitions in the same debate when they address very similar issues, and so the main data of interest come from 84 debates. As discussed below, I also require debate transcripts for all *non-petition* debates during the same parliamentary term. The total corpus consists of 5493 debates and close to 350,000 speeches.[5]

To construct the independent variable, I take the population normalised *constituency*-level signature counts for a given petition, and divide by the population normalised *national*-level signature counts for that petition.[6] The signature rate for MP i in debate d is given by:

$$\text{Signature rate}_{id} = \frac{\text{Constituency Signatures Per Capita}_{id}}{\text{National Signatures Per Capita}_{d}} - 1 \tag{1}$$

where I subtract 1 so that for all non-petition debates the signature rate can be set to 0 for all MPs, meaning that the non-petition debates will not contribute to the estimation of coefficients relating to the signature rate variable. Normalising the counts by population size accounts for the fact that some constituencies are larger than others. Dividing by the national signature rate accounts for the fact that some petitions are more popular than others. Equation 1 therefore implies that an MP's signature rate will be larger when her constituents sign a petition at a greater rate than constituents from other constituencies.

The measure assumes that MPs are more likely to participate in a petition debate if their constituency is high-ranked relative to *other constituencies* in terms of signatures on a given petition. Another plausible assumption is that MPs are more likely to participate in debate on a given petition if they receive more constituency signatures than received for *other petitions*. The former of these possibilities seems most likely. The Petitions Committee provides MPs with constituency signature counts in a spreadsheet format on a petition by petition basis. Making comparisons relative to other constituencies on a given petition is straightforward: an MP (or her staff) need simply look up this information in the spreadsheet.

[5] Petition data was collected from www.petition.parliament.uk and speech data from www.hansard.parliament.uk.

[6] When more than one petition is discussed in a debate, I sum the constituency signature counts across the petitions.

Making comparisons to past petitions is more difficult: an MP would need to analyse historical data to establish whether constituents found the current petition more salient than previous petitions.[7]

In the main analysis, the dependent variable measures whether MP i participates in debate d or not. An MP is coded as participating when she makes a speech in the debate ($y_{id} = 1$), and zero otherwise. Of the 3.5 million MP-debate observations from both petition and non-petition debates, approximately 3% indicate participation. I also consider the effects of signatures on petition *agreement*. Restricting the sample to petition debates, an MP is coded as agreeing with a petition if she expressed overt support for the petition proposal in her speech.[8] Conditional on participating in a petition debate, approximately 65% of MPs agreed with the aims of the relevant petition.

Figure 1 shows the probability of debate participation (panel A) and petition agreement (panel B) over the range of the signature rate variable for all petition debates.[9] Grey points are the raw data and the black lines are a spline fit to the data. The plots illustrate that, in the raw data at least, there is a clear positive relationship between petition signatures and both outcome variables. Bivariate logistic regressions imply that increasing the signature rate by one unit increases the odds of debate participation by 37% (z = 16.28, assuming independence) and, conditional on an MP participating in the relevant debate, the odds of expressing agreement with the petition by 53% (z = 6.56, assuming independence).[10]

Empirical strategy

Simple comparisons of participation between MPs from constituencies with low and high signature rates are likely to be confounded in a number of ways. First, some constituencies will have higher signature counts, on average, than other constituencies.[11] If MPs from these

[7] In appendix section S6, I also show that the main effects presented below are robust to several alternative specifications of the independent variable.

[8] Full details are given below.

[9] Very few observations have a signature rate greater than 20 and in these plots I restrict to values between -1 and 20.

[10] See tables S3 and S4 in appendix section S2.

[11] See appendix section S3 which documents the variation in average numbers of petition signatures across different constituencies.

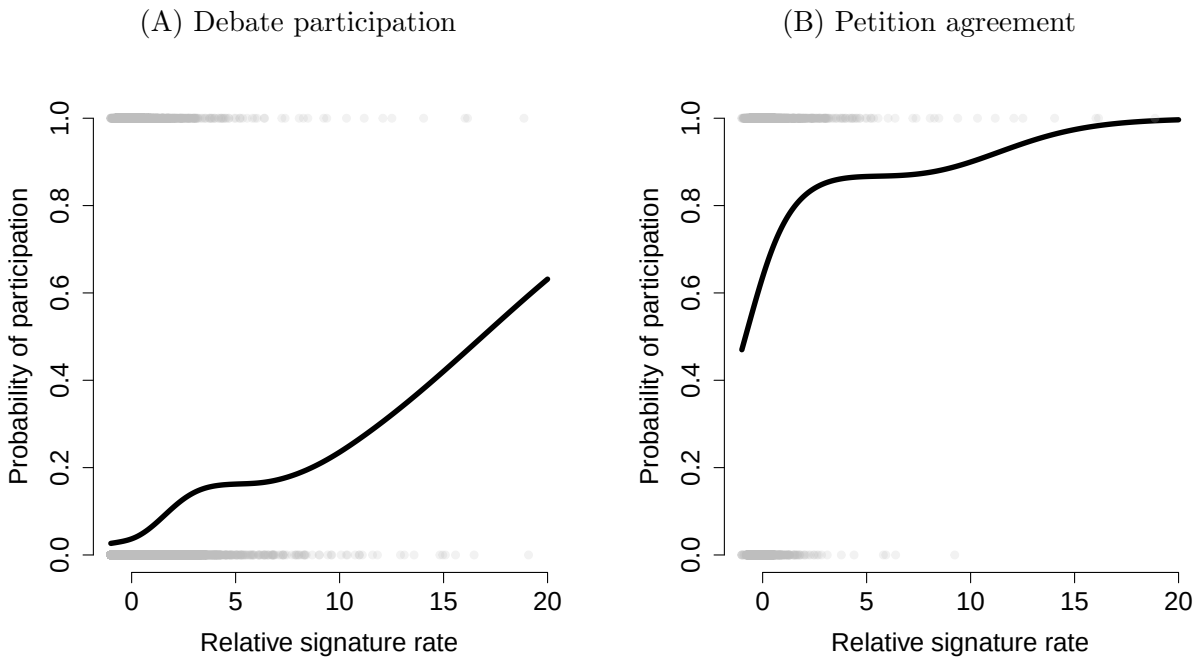

Figure 1: **Petition signatures predict debate participation and petition agreement** *Panel A*: as the petition signature rate increases, MPs are more likely to participate in the relevant petition debate. *Panel B*: conditional on participation, an MP is more likely to express agreement with a petition as the signature rate increases.

constituencies are also more active in debates *in general* than other MPs, then any association between signatures and participation might reflect these general patterns. To overcome this problem I estimate MP-specific intercepts that account for fixed tendencies for some MPs to participate in debate more than others, and which imply that identification of the signature rate effect relies only on within-MP variation.

Second, some petitions are clearly more salient than others. For example, the "EU Referendum Rules triggering a 2nd EU Referendum" petition[12] received more than 4 million signatures nationally and was subject to extensive media interest. Debates on such popular petitions are likely to be better attended on average than other petition debates, and MPs' participation is likely due to the high-profile nature of the debate in general, rather than because of the advocacy of their constituents in particular. I account for this concern in two ways. First, as explained above, the signature rate variable increases only when a constituency provides more signatures for a given petition relative to the total national signature rate for

[12] https://petition.parliament.uk/archived/petitions/131215

that petition. Second, I also include debate-specific intercepts in all models that account for commons shocks to participation that affect all MPs in a given debate.

A more subtle inferential concern, however, is that debate participation may not result from constituent activism via e-petitions, but may instead reflect MPs' pre-existing tendencies to devote attention to the types of issue that attract the attention of their constituents. Consider for example the MP for Brighton Pavillion, Caroline Lucas. Brighton Pavillion is one of the most pro-European constituencies in the UK (Hanretty & Vivyan 2015) and in the "2nd EU Referendum" petition, it ranked 7th (out of 650 constituencies) in terms of the signature rate variable. Caroline Lucas was an active participant in the debate relating to this petition, but she also contributed to many *other* debates that centred on the topic of the UK's relationship with the EU even before this particular petition gained traction. Between May 2015 and June 2016, Ms Lucas mentioned the European Union 77 times in debate: three times as often as the average MP. Even in the absence of the petition, then, it is more likely that Ms Lucas would contribute to another debate on the issue of the EU than would other MPs.

Figure 2 gives two further examples of this type of issue-specific confounding. In both panels, the x-axis indicates the signature rate for an example petition. Panel A depicts a petition calling for the government to increase levels of support for refugees, which was debated in September 2015.[13] Panel B shows a petition which called on the government to prevent the UK's disorderly exit from the EU, which was debated in November 2018.[14] The y-axis measures whether an MP participated in *non-petition* debates that were closely related to these petition debates, and which were held *before* the relevant petition was discussed in parliament. For example, the y-axis in panel A notes whether an MP participated in three debates held throughout mid-2015 which focussed on the refugee crisis. For panel B, the y-axis measures whether an MP participated in three non-petition debates that focussed on the Brexit process.[15] In both cases, petition signatures clearly predict MP participation in

[13] https://petition.parliament.uk/archived/petitions/105991

[14] https://petition.parliament.uk/petitions/219905

[15] I describe the procedure for selecting the three closest matching debates to each petition debate below. The refugee petition is paired with debates on "Refugee Situation in the Mediterranean"; "Calais"; and "Syria: Refugees and Counter-terrorism". For the Brexit petition, the debate is paired with non petition debates on "UK-EU Renegotiation"; "European Union (Notification of Withdrawal) Bill"; and "Leaving the EU: Negotiations".

(A) Participation in non-petition refugee debates (B) Participation in non-petition EU debates

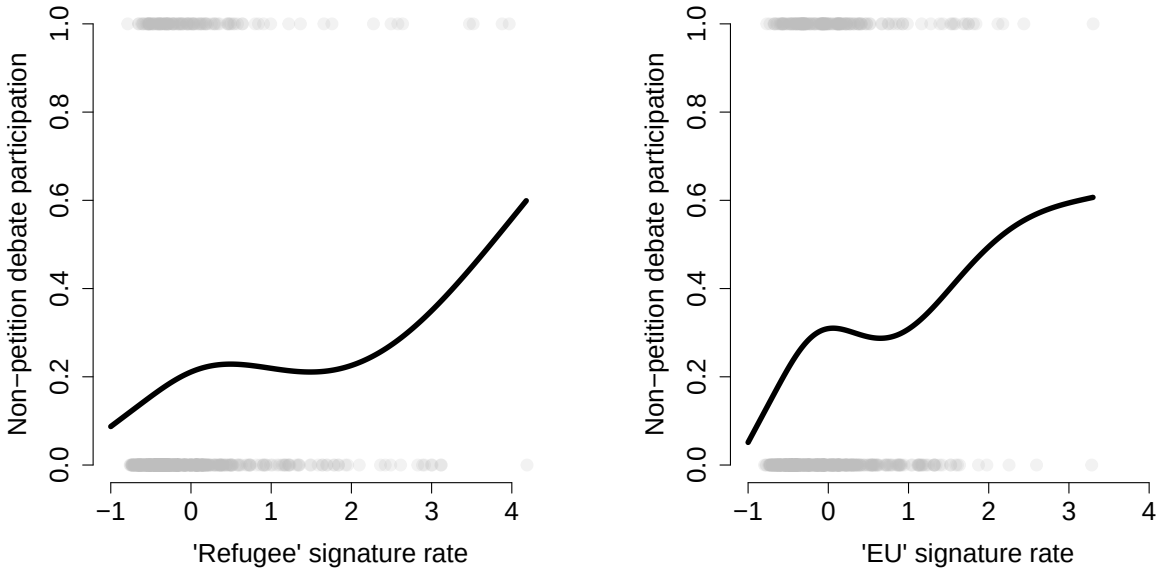

Figure 2: **Petition signatures predict participation in related *non-petition* debates** Panel A: MPs who received more 'Refugee' petition signatures are more likely to participate in non-petition debates pertaining to the refugee crisis. Panel B: MPs who received more 'EU' petition signatures are are more likely to have participated in non-petition Brexit debates.

prior *non-petition* debates that relate to the same issues as those raised in petitions.

Consequently, the relationship between petition signatures and participation is likely confounded by the fact that MPs will tend to speak frequently about issues which are of concern to their constituents, even in the absence of the information that they glean from the petition. This implies that we should account for an MP's general tendency to participate in debates that relate to the issues of interest that are raised in petitions. In essence, we would like to be able to compare an MP's participation on a given petition debate to a counterfactual debate for which the MP was not provided with petition information. Although we cannot observe this counterfactual, we can make comparisons between petition and non-petition debates that deal with similar underlying topics. The empirical challenge is to design a procedure that maps petition debates to other debates in terms of their shared topical content, and quantifies the tendency of each MP to participate in debates on those topics.

Model specification

The empirical strategy proceeds in two steps. First, I use statistical topic models to characterise the topics that feature in parliamentary debate, and to estimate the proportion of each debate that pertains to each topic. Second, I incorporate these debate-specific topic proportions into the participation model by including coefficients for each MP-topic combination. These effects describe the relative tendency for a given MP to participate on a given topic (for both petition and non-petition debates), and allow me to control for the possibility that some MPs will be more prone than others to discuss certain topics. If an MP tends to participate in many debates about healthcare, for example, then that MP will have a large 'healthcare' topic random-effect. To the extent that an MP's participation in debates relevant to specific topics correlates with her constituents' signatures of petitions that are also relevant to those topics, this strategy will lead to more conservative estimates of the effects of petition signatures.

To implement the topic-control strategy, I estimate statistical topic models using the texts of all speeches during the study period. In particular, I concatenate all speeches within each debate into a common text, and then estimate Correlated Topic Models (Blei & Lafferty 2005) models for all D parliamentary debates, including petition debates.[16] Like all topic models, the CTM assumes that the frequency with which terms co-occur within different debates gives information about the 'topics' that feature in those debates. The key output of the model is λ, which is a $K * D$ matrix, where λ_d describes a given debate as a vector of topic proportions, and λ_{kd} gives the proportion of debate d devoted to topic k. Note that I use the texts of *all* debates – both petition-related and non-petition-related – to estimate the topic models. This is essential, as the resulting λ matrices therefore provide a common topic space which we can use to compare petition debates to regular debates.

The CTM requires that the we choose a number of topics, K, to estimate from the data. Because there is no *a priori* reason to prefer any particular value for K, I estimate several different models, while varying the number of topics: $K \in 5, 10, ..., 100$. I present results from all 20 topic model specifications in appendix figure S2, but appendix section S5 shows that

[16] I remove all punctuation and numbers from the texts, reduce words to their stems, and remove any tokens that appear in fewer than 2.5% of all texts. I implement the CTM as the null specification of the structural topic model (Roberts et al. 2013)

the model with 45 topics best predicts MP debate participation and I focus on the results from this model for the remainder of the paper.

The validity of the control strategy rests on the assumption that we can approximate the topics of petition debates in the full sample of non-petition debates. To investigate this assumption, in appendix table S8, I present for each petition debate in the data, the titles of the three closest matching *non-petition* debates according to the cosine-similarity between the λ_d vectors. The results are reassuring, as it is clearly the case that petitions debates are matched with non-petition debates that deal with very similar policy areas. For example, the petition debate on "Student Loan Agreements" is matched to non-petition debates on "Higher Education Funding", "Student Maintenance Grants", and "Funding Higher Education". Similarly, the petition debate on "Fur Trade" is matched to non-petition debates on "Foie Gras Imports", "Animal Welfare", and "Puppy Smuggling". Across all debates in the sample, the topic model clearly clusters petition debates with non-petition debates that share a very similar substantive focus.[17]

With the topic estimates in hand, for individuals $i, ..., N$, debates $d, ..., D$, and topics $k, ..., K$, I model debate participation in a hierarchical logistic regression of the following form:

$$\text{logit}(y_{id}) = \alpha_0 + \alpha_i + \delta_d + \sum_{k=1}^{K} \beta_{ik}\lambda_{kd} + \gamma_{i,t} * \text{Signature rate}_{id} \tag{2}$$

where y_{id} measures whether an MP participated in a given debate and Signature rate$_{id}$ is defined in equation 1. α_i and δ_d are intercepts for MPs and debates, respectively, α_0 is a grand-intercept term, and β is a matrix of MP-topic coefficients. Finally, γ is a vector of MP-signature coefficients.

The primary quantities of interest are the $\gamma_{i,t}$ coefficients, which describe the relationship between the signature rate and debate participation for each MP in each parliamentary session for which they appear in the data.[18] As these coefficients are estimated for each MP, I use

[17] In appendix section S5 I provide additional validation of the topic control strategy. In particular, I show that this approach improves our ability to predict MP debate participation in general and also associates MPs to topics in ways that are consistent with intuitions about British politics.

[18] I include separate coefficients for each MP in each term because I model these signature effects as a

only within-MP variation in debate participation to identify the effect of petition signatures. Recall also that as Signature rate$_{id} = 0$ for all non-petition debates, it is only petition debates that contribute to the likelihood function via the γ parameters.

As outlined above, I also expect the effect of signatures to vary across different MP-types, and I model the γ coefficients in a second-level model of the following form:

$$\begin{aligned} \gamma_{i,t} \quad \sim \quad & N(\varphi_0 + \varphi_{\text{New MP}} \cdot \text{New MP}_i + \\ & \varphi_{\text{Frontbench}} \cdot \text{Frontbench}_{i,t} + \\ & \varphi_{\text{Margin}} \cdot \text{Margin}_{i,t} + \\ & \sum_{p}^{P} \varphi_{\text{Party}_p} \cdot \text{Party}_{i,p}, \sigma_{\gamma}^2) \end{aligned} \tag{3}$$

where Frontbench$_{i,t}$ is a binary variable for whether a given MP held a frontbench position for either the government or the main opposition party (Labour) during a given parliamentary session, New MP$_{i,t}$ is an indicator for whether the MP was new to the House in the parliamentary session t, and Margin$_{i,t}$ is the electoral margin of victory for MP i in the election preceding time period t. I also control for the party of MP i via the Party$_{i,p}$ dummy variables, though the results are not sensitive to this choice.[19] The coefficients associated with these second-level variables will tell us how (on the logit scale) the γ_i coefficients vary, on average, according to the attributes of the MP.

The model provides two major benefits. First, it represents a conservative strategy for estimating the effect of signatures on participation. MP-specific intercepts rule out omitted variable bias that stems from MP characteristics that are time invariant; debate-specific intercepts control for confounding factors that relate to the aggregate popularity of each debate; and the MP-topic coefficients account for any factors that might drive MP and constituent attention to particular topics. Second, it provides a rich framework for evaluating how the effects of e-petitions vary as a function of MP-level characteristics.[20]

function of covariates that change between parliamentary terms. Most MPs served in both the 2015-17 and 2017-Present parliaments, and so these MPs have two associated γ coefficients. 96 MPs served only in the 2015-17 parliament, and 94 served only in the 2017-Present parliament. The effect of signatures on participation are described by a single coefficient for each of these MPs.

[19] See table S13 in the appendix.

[20] Although of less substantive interest, I also include the same covariates in a second-level model for the

Estimation is complicated by the fact that we have lots of data (≈ 3.5 million observations), lots of models ($K \in 5, 10, ..., 100$), and lots of parameters per model.[21] Traditional Bayesian inference approaches such as MCMC would be computationally impossible here, and so I instead use a variational inference approach (Kucukelbir et al. 2017) to approximate the posterior distribution, which I implement in Stan (Carpenter et al. 2016).[22]

Results

Table 1 presents median estimates and associated 95% posterior intervals for the second-level logit coefficients described in equation 3.[23] Model 1 includes only the γ parameters from equation 2 and the associated second-level parameters from equation 3. Models 2 and 3 include intercepts for MPs (α) and debates (δ), respectively, and both sets of intercepts are included in model 4. Model 5 additionally includes the MP-topic coefficients (β). Although there are some small differences across models, conditioning on MP-effects, debate-effects, and MP-topic-effects makes little substantive difference to the estimated effects of petition signatures. That the signature-rate effects differ little after the inclusion of these controls is reassuring, and considerably strengthens the inferences drawn here.

Consider the baseline signature effect in each model, φ_0, which represents the effect of the

MP-intercepts ($\alpha_{i,t}$), and model debate effects as a function of whether a debate is associated to a petition or not. The model is completed by uninformative prior distributions over the β_{ik} coefficients, and the variance parameters:

$$\begin{aligned}
\alpha_{i,t} &\sim N(\mu_0 + \mu_{\text{New MP}} * \text{New MP}_{i,t} + \\
&\quad \mu_{\text{Frontbench}} * \text{Frontbench}_{i,t} + \\
&\quad \mu_{\text{Margin}} * \text{Margin}_{i,t} + \\
&\quad \sum_{p}^{P} \mu_{\text{Party}_p} \cdot \text{Party}_{i,p}, \sigma^2_\alpha) \qquad (4) \\
\delta_d &\sim N(\phi_0 + \phi_{\text{Petition Debate}} * \text{Petition Debate}_d, \sigma^2_\delta) \\
\beta_{i,k} &\sim N(0, \sigma^2_{\beta_k}) \\
\sigma_\alpha, \sigma_\delta, \sigma_\gamma, \sigma_{\beta_k} &\sim N(0, 2)
\end{aligned}$$

[21] A model with 45 topics includes over 40 thousand parameters, and a model with 100 topics includes over 80 thousand.

[22] Variational inference is orders of magnitude faster than state-of-the-art MCMC algorithms, and is guaranteed to provide the correct expected values of parameters. However, it will also sometimes understate the variability in the posterior – particularly when parameters are correlated – implying that in some cases the estimated standard errors may be too small.

[23] Estimates for the additional coefficients described in equation 4 are given in appendix table S11.

	Model 1	Model 2	Model 3	Model 4	Model 5
$\varphi_{\textbf{Margin}}$	-0.69 (-0.95, -0.45)	-0.39 (-0.63, -0.15)	-0.52 (-0.78, -0.23)	-0.48 (-0.69, -0.26)	-0.38 (-0.69, -0.09)
$\varphi_{\textbf{Frontbench}}$	-0.12 (-0.32, 0.1)	-0.15 (-0.42, 0.1)	-0.06 (-0.23, 0.11)	-0.11 (-0.28, 0.08)	-0.36 (-0.53, -0.18)
$\varphi_{\textbf{New MP}}$	0.29 (0.14, 0.42)	0.37 (0.24, 0.51)	0.29 (0.17, 0.39)	0.33 (0.25, 0.42)	0.24 (0.14, 0.33)
φ_0	0.14 (0.08, 0.2)	0.02 (-0.05, 0.08)	0.25 (0.19, 0.31)	0.31 (0.25, 0.38)	0.2 (0.13, 0.27)
φ Party dummies	✓	✓	✓	✓	✓
MP random effects	×	✓	×	✓	✓
Debate random effects	×	×	✓	✓	✓
MP-topic random effects	×	×	×	×	✓
N parameters	1276	2549	6772	8045	40901
DIC	889976.4	841711	845104.3	793486.7	726863.5

Table 1: **Second-level φ effects (debate participation)**
Median estimates and associated 95% posterior intervals for the second-level φ coefficients described in equation 3.

petition signature rate for a Conservative MP in a perfectly competitive constituency (where Margin = 0), who is not new to the House and who does not hold a frontbench position. Across all specifications, this baseline effect is positive, and precisely estimated. Based on model 5, increasing the signature rate by one unit is associated with a 22% [14%, 31%] increase in the odds of debate participation for an MP of this type.

To further explore the heterogeneity in these effects, I plot the second-level φ coefficients for model 5, with associated 90 and 95% posterior intervals, in panel A of figure 3. The plot reveals important interaction effects between MP characteristics and the signature-rate. For example, in line with hypothesis 4, the estimated $\varphi_{\text{Frontbench}}$ coefficient is negative, implying that the effect of petition signatures is smaller for MPs who hold government or opposition frontbench positions than it is for backbench MPs. Similarly, and consistent with hypothesis 3, the $\varphi_{\text{New MP}}$ coefficient is positive: MPs who are new to the house, and therefore have had less time to establish a reputation for constituency service, are significantly more responsive to petition signatures than are more experienced MPs.

The negative coefficient for φ_{Margin} also provides support for hypothesis 2, and suggests that the effect of petition signatures is larger for MPs from more competitive constituencies.

(A) Second-level (φ) coefficients (B) Competitiveness and participation

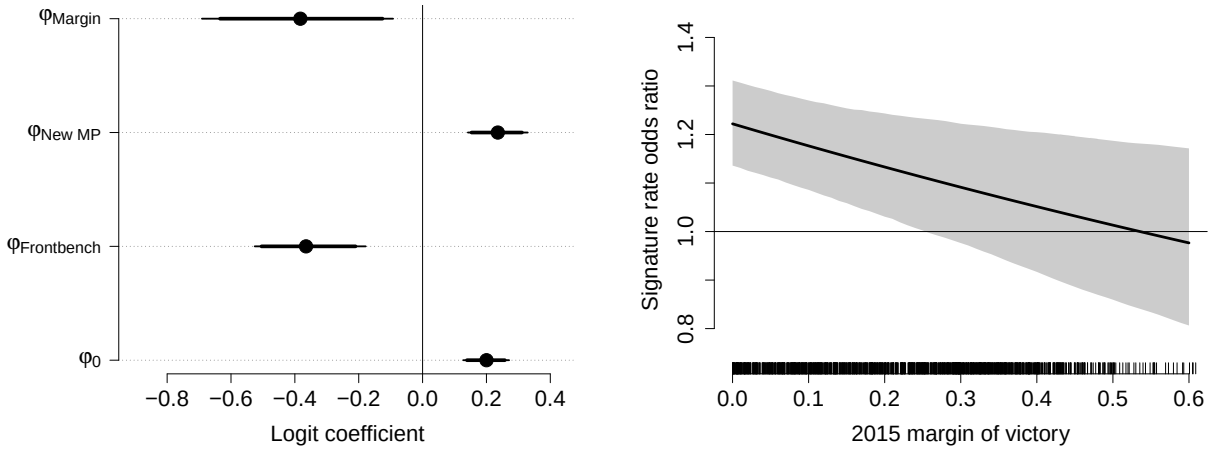

Figure 3: **Effects of signature rate on debate participation**
Panel A: median estimates and 90% (thin lines) and 95% (thick lines) posterior intervals for the φ coefficients from model 5 in table 1.

Panel B of figure 3 depicts the effects of the signature-rate on participation (as an odds-ratio) across the range of the Margin variable. The figure shows that the effect of petition signatures is strongest for MPs in competitive constituencies, and weakest for MPs in very safe seats. The substantive magnitude of this effect is non-trivial: increasing the signature rate in a competitive constituency (Margin = 1) from half of the national rate to 10 times the national rate leads to a 5 percentage point increase in the probability of participating in debate,[24] relative to a baseline probability of just 3 percent. However, the same increase in the signature rate for uncompetitive constituencies (Margin = 40) results in no increase in participation probability.

In sum, the results here suggest that even conditioning on the topics under consideration, MPs are responsive to constituent activism as expressed through e-petitions. The model also reveals that this responsiveness is clearly conditioned by party discipline considerations, and by the electoral incentives an MP faces. Overall, however, in terms of encouraging legislators to engage with the issues that their constituents find most salient, e-petitions appear to have positive and non-trivial effects on dyadic representation.

[24]Based on Conservative MPs who do not hold a frontbench position and who are not new to the house.

Petition agreement

For those MPs who do participate in petition debates, is there a relationship between signatures and petition *agreement*? If e-petitions are effective, we would expect MPs who receive many signatures from their constituents to actively support a petition during the course of debate. I investigate this question by focussing on the subset of 581 MPs who participated in the 84 petition debates in the sample. An MP is coded as being in agreement with a petition if she expressed overt support for the proposal given in the petition text.[25] In this section, the dependent variable therefore measures whether an MP expressed agreement with a given e-petition ($y_{id} = 1$) or not ($y_{id} = 0$).

As with the previous analysis, I model MP agreement using a hierarchical logit model which includes MP-specific signature effects (γ_i), MP intercepts (α_i), and debate intercepts (δ_d) at the first level. Here, the MP effects capture the tendency for MPs to agree with petitions in general, and the debate effects capture whether some petitions are more agreeable to all MPs than others. Predictors at the second-level of the model are identical to those given in equation 3: I model the MP-signature effects (γ_i) as a function of whether the MP is new to the House, whether the MP held a frontbench position, and the electoral margin of the MP.[26]

In contrast to the analysis above, I am unable to estimate the MP-topic random effects (β_{ik}) here. To do so would require coding each MP's *position*, rather than merely their participation, on each of the non-petition debates discussed in parliament. Manual coding of this sort is infeasible for such a large number of debates, and traditional scaling approaches are not well-suited to a system with strong party discipline and strategic voting (Spirling & McLean 2006). Accordingly, because this secondary analysis is potentially prone to MP-specific topic confounding, the results should be considered as illustrative.

[25] In cases where more than one petition was discussed in a given debate, coding was based on agreement with the largest petition. To assess coder reliability, the author and an RA read 1384 petition debate speeches and coded whether each MP was "For" or "Against" the motion in the petition. Coders agreed in 90 % of cases.

[26] Again, I also control for the party of the MP and I include the same predictors in the second-level model for the α_i parameters (appendix table S12).

	Model 1	Model 2	Model 3	Model 4
$\varphi_{\textbf{Margin}}$	0.29 (-0.27, 0.8)	0.41 (-0.21, 1.04)	0.82 (-0.03, 1.61)	0.19 (-0.52, 0.92)
$\varphi_{\textbf{Frontbench}}$	-0.05 (-0.78, 0.79)	-0.37 (-1.15, 0.4)	0.12 (-0.86, 1.14)	0.11 (-1.13, 1.26)
$\varphi_{\textbf{New MP}}$	0.05 (-0.17, 0.27)	0.09 (-0.1, 0.29)	0.45 (0.21, 0.71)	0.55 (0.25, 0.85)
φ_0	0.95 (0.75, 1.14)	1.02 (0.83, 1.21)	0.58 (0.35, 0.8)	1.03 (0.83, 1.25)
φ Party dummies	✓	✓	✓	✓
MP random effects	×	✓	×	✓
Debate random effects	×	×	✓	✓
N parameters	796	1589	882	1675
DIC	4046.1	3636.3	2672.4	2698.3

Table 2: **Second-level φ effects (petition approval)**
Median estimates and associated 95% posterior intervals for the second-level φ coefficients for the petition approval model.

Results

The estimates for the second-level φ coefficients from the approval analysis are presented in table 2. As before, the estimates of the second-level coefficients are relatively stable across the different specifications, though, due to the much smaller sample size, they are also somewhat noisier.

In general, this analysis suggests that in addition to the effects on debate participation, petition signatures also matter for MP approval. The baseline signature-effect parameter, φ_0, is positive and significantly different from zero in all model specifications. Based on the estimates in model 4, for a non-frontbench Conservative MP who is not new to the house, and has the mean level of electoral competitiveness, increasing the signature rate from half of the national rate to 10 times the national rate is associated with a 67 percentage point increase in the probability of petition approval. Conditional on participating in petition debate, MPs are substantially more likely to express support for a petition in their speeches when many of their constituents have signed that petition. While these approval results are potentially subject to the topic-confounding concerns explained in previous sections, the evidence is broadly consistent with the notion that MPs use petition debates to demonstrate responsiveness to

constituent concerns.

Consistent both with hypothesis 3 and the participation results presented above, MPs who have recently joined the house are more sensitive to increases in the petition signature rate than are more experienced MPs. There is, however, less evidence of heterogeneity here. For instance, there is no predictable variation in the effect of petitions signatures on petition agreement either as a function of electoral margin, or whether an MP holds a frontbench position. The estimates on these second-level variables are also considerably less precise than in the participation analysis.

What accounts for these differences between the participation and agreement analyses? One key point is that the baseline level of petition agreement is very high. Of the 1994 speeches in the raw data, 1302 expressed agreement with the aims of the petition. Furthermore, as is clearly visible in panel B of figure 1, even MPs from constituencies with low signature counts are – conditional on participating – more likely than not to express agreement with the petitions in debate. These patterns suggest that the main representational decision from an MP's perspective is whether or not to participate in certain debates, rather than which position to take in the debates in which they do participate. Nevertheless, the main results here are consistent with the participation results presented above: e-petitions provide useful information about constituent preferences, and this information can strengthen the dyadic relationship between voters and MPs.

Conclusion

On June 24th 2016, MPs in Westminster Hall debated a petition calling on the government to "Prevent Donald Trump from making a State Visit to the United Kingdom" which received nearly 2 million signatures. The debate was very well attended, so much so that the sitting was extended to accommodate the increased demand for floor time. Although the government opposed the petition's aims, MPs from across the political spectrum spoke in favour of cancelling the state visit. In the final speech of the debate, Paul Flynn MP concluded: "This has been an extraordinary event... We are expressing the voice of the people and a thunderous voice it has been... It is a good day for Parliament." (Flynn 2017) The main

findings here suggest that this is more than simple hyperbole. In general, when constituent opinion as expressed via an e-petition is strong, MPs are more likely to lend their voice to the articulation of that opinion in parliamentary debates. These results are therefore encouraging, as they suggest that providing voters with new mechanisms for communicating their political preferences can lead to stronger representational ties between citizens and politicians.

Importantly, the results here suggest that e-petitions add something to pre-existing representational links that connect MPs to their constituents. MPs will naturally speak frequently about issues which are of concern to their constituents even if they are not directly called upon to do so by a petition. However, the empirical approach I use implies that, when a petition is signed by many local people, the MP is more likely to participate in the relevant debate than we would expect given the topical content of the debate alone. This suggests that e-petitions do indeed contain valuable information, and that politicians make use of this information when deciding how to allocate time to different issues in parliament.

The conditional nature of these effects should, however, highlight the limits of policy innovations of this type. Notably, the representational benefits of e-petitions are clearly stronger for some types of MPs. Perhaps unsurprisingly, e-petitions have essentially no effect on MPs for whom party discipline binds most strongly: frontbench MPs do not appear to adapt their behaviour in response to local petition support. The effectiveness of online activism is also clearly conditioned by incentives created by the electoral system: new MPs and MPs from marginal seats are most sensitive to variation in constituent issue priorities. Accordingly, while some see the internet as a force which has the potential to dramatically alter representational relationships in politics (Coleman & Blumler 2009), the findings here suggest that online political activism is more likely to shape political behaviour at the margins and will likely be constrained by existing political incentives.

One potential concern with the results presented above is that rather than e-petitions being a tool for constituents to communicate their preferences to legislators, they may instead be a tool for legislators to mobilise public support for their own favoured projects. However, there are two reasons why this reverse-causality argument is not a serious concern here. First, in contrast to traditional petitions in the Commons, e-petitions do not require the support or

'sponsorship' of an MP before they are considered for debate, and previous work shows that e-petitions are normally promoted by individual citizens with little help from MPs themselves (Wright 2015*b*). Second, the vast majority of traffic to the website that hosts e-petitions comes via social media (Yasseri et al. 2017, 11). If MPs encourage their constituents to sign particular petitions, this should be obvious from their activity on social media platforms. In appendix section S7, I show that very few MPs use social media to promote e-petitions, making it unlikely that the results presented above are attributable to MP activism.

These findings also speak to a growing empirical literature on dyadic representation in parliamentary systems. The combination of an SMD electoral system and a parliamentary government creates countervailing incentives for MPs: be a loyal party member and improve the party brand, or respond faithfully to constituent demands and increase electoral security by building a strong personal vote. While party discipline is clearly a primary determinant of MPs' *voting* decisions in these systems, the findings here highlight that responsiveness to voter preferences via other forms of legislative behaviour is non-zero. Whether they navigate the party-voter trade off by making use of roll-call votes that are not subject to party discipline (Hanretty et al. 2016), by internal lobbying of party leaders (Butler et al. 2017), or, as demonstrated here, through the strategic use of parliamentary speech (Soroka et al. 2009, Blidook & Kerby 2011), it is clear that MPs are aware of and respond to constituent preferences with many different types of legislative behaviour.

There are, of course, many other dimensions on which we might want to evaluate the effectiveness of online activism. A common criticism of e-petition systems, for example, is that even popular petitions have little tangible impact on the policy-making process (Wright 2015*a*). For instance, despite the vocal opposition of many MPs in the Trump debate, the US President did indeed visit the UK in mid-2018. However, there are some instances where e-petitions clearly played a role in promoting policy change (Miller 2008), and more work is need to assess what (if any) downstream consequences the behavioural effects documented here have for policymaking. Future research might also address whether e-petitions help cultivate further interest in and understanding of politics, and whether they encourage signatories to participate in politics in other ways.

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Appendix

S1. Summary statistics

Table S1 presents descriptive statistics for the main variables used in the participation analysis, and tables S2 gives the equivalent statistics for the approval analysis.

Table S1: Summary statistics

Statistic	Mean	Median	Max	Min	St. Dev.	N
Participate	0.03	0	1	0	0.17	3,450,612
Signature Rate	0.0001	0	202	−1	0.15	3,450,612
Margin	0.24	0.24	0.77	0.0000	0.15	3,450,612
Frontbench	0.08	0	1	0	0.27	3,450,612
New MP	0.34	0	1	0	0.47	3,450,612

Table S2: Summary statistics

Statistic	Mean	Median	Max	Min	St. Dev.	N
Approve	0.65	1	1	0	0.48	1,994
Signature Rate	0.50	0.02	201.92	−1.00	5.02	1,994
Margin	0.22	0.19	0.77	0.001	0.15	1,994
Frontbench	0.02	0	1	0	0.12	1,994
New MP	0.50	0	1	0	0.50	1,994

S2. Bivariate logistic regression results

Table S3: Bivariate logit – Participation

	Participation
Relative Signature Rate	0.313
	(0.019)
Constant	−3.213
	(0.023)
Observations	52,733
Akaike Inf. Crit.	17,435.510

Note:

Table S4: Bivariate logit – Approval

	Approval
Relative Signature Rate	0.427
	(0.065)
Constant	0.542
	(0.048)
Observations	1,992
Akaike Inf. Crit.	2,510.917

Note:

S3. Descriptive analysis of petition signatures

Table S5 presents the results for a simple descriptive analysis which documents the variation in aggregate petition signature rates across parliamentary constituencies in the UK. I calculate the total number of signatures per constituency across all petitions in the data that received over 1000 signatures nationally. I then divide this by the number of people living in the constituency in 2015 to get the aggregate number of signatures per capita, and use this as an outcome variable in a linear regression model. The right-hand side variables are demographic – median pay; percentage female; percentage under 30; population density; percentage of students; percentage working in industry; percentage of home owners; percentage white; percentage unemployed – or political – vote for 'Leave' in the 2016 UK-EU referendum; Labour, Liberal Democrat, SNP, and 'Other' vote share in 2015. In model 2 of table S5, I also include regional fixed-effects.

The table shows that the volume of signatures that constituencies contribute across petitions vary in predictable ways. Constituencies which have high numbers of young people, lower levels of industrial workers, lower levels of unemployment and which are less ethnically diverse tend to contribute more signatures to e-petitions. Similarly, constituencies in which more people supported 'Remain' in the EU referendum are significantly more likely to sign e-petitions. These findings are consistent on petition-signing in Germany, where Lindner & Riehm (2011), for example, find that individuals who sign e-petitions are younger and more educated, on average, than the general population.

Table S5: Constituency-level determinants of signature rates

	Signatures per capita	
	(1)	(2)
Median pay	0.0003	−0.001
	(0.001)	(0.001)
Female	0.004	−0.001
	(0.005)	(0.005)
Young	0.007***	0.007***
	(0.002)	(0.002)
Pop. Density	0.001***	0.001**
	(0.0003)	(0.0003)
Students	−0.004**	−0.004**
	(0.002)	(0.002)
Industry	−0.007***	−0.005***
	(0.001)	(0.001)
Home Ownership	−0.00003	0.001
	(0.001)	(0.001)
White	0.003***	0.004***
	(0.0003)	(0.0003)
Unemployed	−0.016***	−0.005
	(0.004)	(0.004)
Leave Share	−0.592***	−0.763***
	(0.051)	(0.050)
Labour 2015	−0.017*	0.012
	(0.009)	(0.009)
LibDem 2015	−0.041	0.001
	(0.026)	(0.024)
SNP 2015	−0.256***	−0.076*
	(0.015)	(0.039)
Other 2015	0.013	0.074**
	(0.033)	(0.030)
Constant	0.613**	0.702***
	(0.266)	(0.233)
Regional Fixed-Effects	No	Yes
Observations	631	631
R^2	0.646	0.736
Note:	*p<0.1; **p<0.05; ***p<0.01	

S4. Debated petitions

Table S6 presents the titles, total signature counts, and debate dates of all 108 debated e-petitions in the data.

Table S6: Debated e-petitions

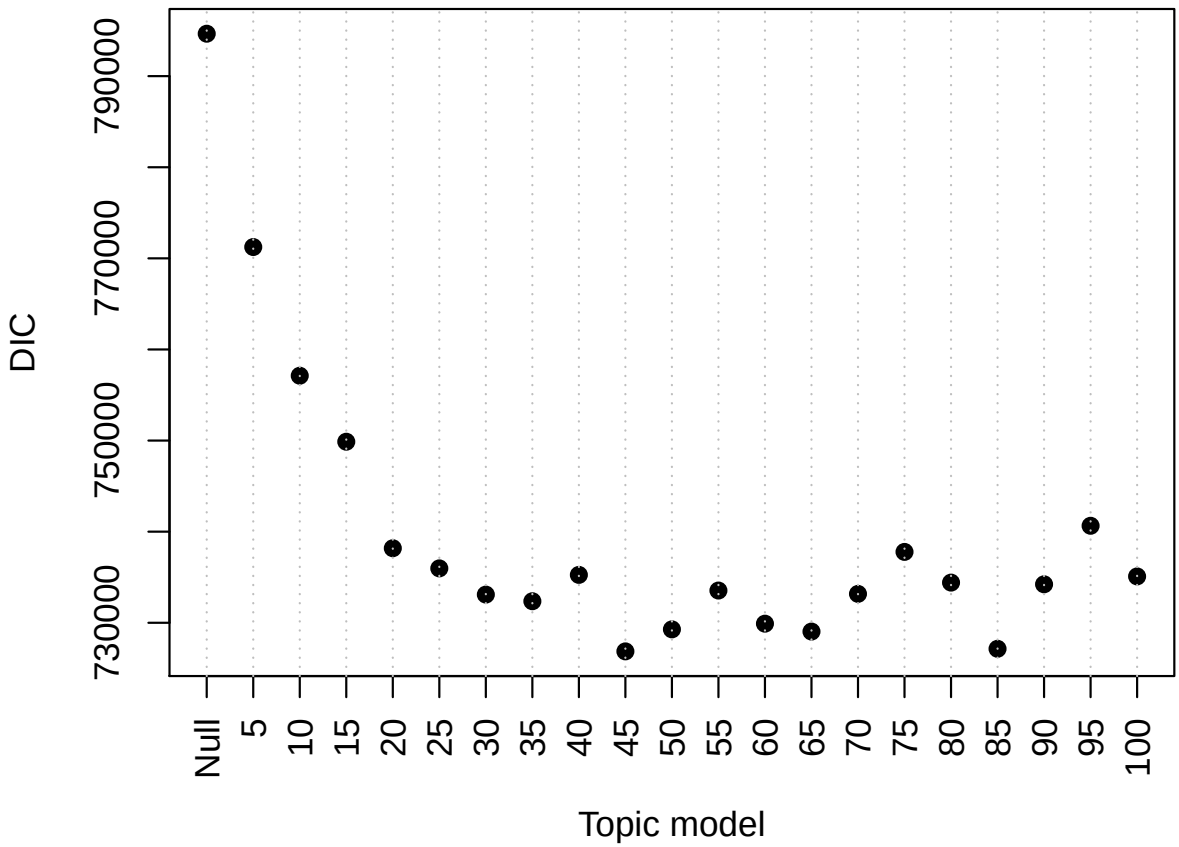
Title	Signatures	Date
EU Referendum Rules triggering a 2nd EU Referendum	4150262	Sep 2016
Prevent Donald Trump from making a State Visit to the United Kingdom.	1863707	Feb 2017
Give the Meningitis B vaccine to ALL children, not just newborn babies.	823349	Apr 2016
Block Donald J Trump from UK entry	586930	Jan 2016
Accept more asylum seekers and increase support for refugee migrants in the UK.	450287	Sep 2015
Leave the EU without a deal in March 2019.	358540	Jan 2019
Donald Trump should make a State Visit to the United Kingdom.	317542	Feb 2017
Ban the sale of fireworks to the public. Displays for licenced venues only.	299785	Nov 2018
Put pressure on Libya to take action to stop enslavement of Black Africans.	269159	Dec 2017
Stop the privatisation of NHS services	238767	Apr 2018
Make the production, sale and use of cannabis legal.	236995	Oct 2015
Stop spending a fixed 0.7 per cent slice of our national wealth on Foreign Aid	235979	Jun 2016
To debate a vote of no confidence in Health Secretary the Right Hon Jeremy Hunt	231136	Sep 2015
Vote no on military action in Syria against IS in response to the Paris attacks	227745	Dec 2015
STOP CAMERON spending British taxpayers money on Pro-EU Referendum leaflets	221866	May 2016
Stop allowing immigrants into the UK.	216949	Oct 2015
Lower the age for smear tests from 25 to 18 to prevent cervical cancer.	205357	Jan 2019
No more school penalty fines and bring back the 10 day authorised absence	204790	Jul 2016
Rescind Art.50 if Vote Leave has broken Electoral Laws regarding 2016 referendum	203230	Sep 2018
Make fair transitional state pension arrangements for 1950s women	193185	Feb 2016
No UK airstrikes on Syria.	190223	Dec 2015
Reject calls to add Staffordshire Bull Terriers to the Dangerous Dogs Act	186226	Jul 2018
Put a max of 1200 on car insurance for 18-25 year olds	185175	Mar 2017
School should start at 10am as teenagers are too tired	184602	Feb 2019
Amnesty for anyone who was a minor that arrived In Britain between 1948 to 1971	180483	Apr 2018
Give the electorate a referendum on the abolition of the House of Lords	170983	Jun 2018
Change the GCSE English Literature exam from closed book to open book.	165934	Mar 2018
Keep the NHS Bursary	162568	Jan 2016
Call on PM to take action to build public trust in the Grenfell Tower Inquiry	156834	May 2018
Introduce a tax on sugary drinks in the UK to improve our children's health.	155516	Nov 2015
Make it illegal for a company to require women to wear high heels at work	152420	Mar 2017
Pay Up Now! Scrap the pay cap and give public servants a meaningful pay rise	149138	Dec 2017
Close all retail on boxing day, retail isn't needed on boxing day!	148327	Dec 2016
Ban the sale of puppies by pet shops and all commercial 3rd party dealers.	148248	May 2018
Hold a referendum on the final Brexit deal	145117	Dec 2017
Leave the EU immediately	141839	Jan 2018
Make TV election debates happen - establish an Independent Debates Commission	141444	Jan 2019
Stop possible second referendum on E.U. membership	135938	Dec 2018
Grant a People's Vote if Parliament rejects the EU Withdrawal Agreement	135602	Jan 2019
Stop retrospective changes to the student loans agreement	134194	Jul 2016
Free childcare when both parents are working. Not just those who are on benefits	133921	Nov 2016
Prevent the scrapping of the maintenance grant.	133069	Jan 2016
Invoke Article 50 of The Lisbon Treaty immediately.	128238	Oct 2016
Require supermarkets to offer a plastic-free option for all their fruit and veg.	128001	Nov 2018
Give status to Police Dogs and Horses as 'Police Officers'	127729	Nov 2016
April's Law	127296	Mar 2017
Make an allowance for up to 2 weeks term time leave from school for holiday.	127199	Oct 2015
Ban driven grouse shooting	123077	Oct 2016
Jeremy Hunt to resume meaningful contract negotiations with the BMA.	120680	Mar 2016
Fund more research into brain tumours, the biggest cancer killer of under-40s	120129	Apr 2016
Keep Childcare Vouchers open beyond April 2018	119588	Jan 2018
Make it a specific criminal offence to attack any member of NHS Medical Staff.	118056	Feb 2017
Make Orkambi available on the NHS for people with Cystic Fibrosis	118033	Mar 2018
Make fair transitional state pension arrangements for 1950s women	117715	Dec 2017
Force child cancer to the forefront of the NHS and government funding schemes.	117183	Nov 2016
Brexit re article 50 it must not be suspended/stopped under any circumstances	116481	Feb 2019
Scrap the 35k threshold for non-EU citizens settling in the UK	114636	Mar 2016
Scrap plans forcing self employed and small business to do 4 tax returns yearly	114504	Jan 2016

Parliament's vote on the Brexit deal must include an option to remain in the EU.	113614	Jun 2018
Change the laws governing the use of fireworks to include a ban on public use	113285	Jan 2018
STOP BREXIT	112576	Jan 2019
Allow Premier League and Championship football clubs to introduce safe standing	112026	Jun 2018
Authorise open book examinations for GCSE English Literature 2017	110876	Apr 2017
If there is no agreement to leave the EU then brexit must be stopped	110134	Nov 2018
The DDRB's proposals to change Junior Doctor's contracts CANNOT go ahead.	110065	Oct 2015
Ban the sale of animal fur in the UK.	109554	Jun 2018
Shut down the domestic ivory market in the UK.	108530	Feb 2017
End the badger cull instead of expanding to new areas	108320	Mar 2017
Give parents the right to opt their child out of Relationship and Sex Education	108202	Feb 2019
Reclassify the theft of a pet to a specific crime in its own right.	107353	Jul 2018
Demand an end to the pay restraint imposed on agenda for change NHS staff.	106409	Jan 2017
Debate in the House the Local Government Pension Scheme Investment Regulations	105772	Oct 2016
Create a new independent welfare body to protect racehorses from abuse and death	105361	Oct 2018
Restrict the use of fireworks to reduce stress and fear in animals and pets	104038	Jun 2016
Include expressive arts subjects in the Ebacc	102499	Jul 2016
Urge the South Korean Government to end the brutal dog meat trade	102131	Sep 2016
Ban all non-recyclable/non-compostable packaging in the UK	101941	Jan 2017
End the export of live farm animals after Brexit.	100752	Feb 2018
Don't kill our bees! Immediately halt the use of Neonicotinoids on crops	99909	Dec 2015
Increase funding for schools	99015	
Ban the sale of products containing unsustainably sourced palm oil in the UK	89802	Dec 2018
Increase college funding to sustainable levels - all students deserve equality!	71257	Jan 2019
Introduce automatic approval of visit visas for families of British Citizens	71178	Jul 2018
Leave the EU now	66973	Jan 2019
Insurance should be on the car itself instead of the individuals who drive it	60447	Mar 2018
Public Holiday on Muslims religious occasions in the UK(Eid ul FitrandEid ul Adha)	52453	Oct 2018
Prevent avoidable deaths by making autism/learning disability training mandatory	52148	Oct 2018
Don't ban Trump from the United Kingdom	46622	Jan 2016
Make British Sign Language part of the National Curriculum	35203	Mar 2018
Free Sergeant Alexander Blackman	34440	Sep 2015
Repeal the 1972 European Communities Act immediately on EU Referendum OUT vote.	32943	Oct 2016
Impose Financial sanctions on the State of Myanmar (Burma).	30349	Apr 2018
Stop Brexit if parliament rejects the deal	29360	Jan 2019
Ask the Government not to invoke article 50 of the Lisbon treaty.	27151	Oct 2016
Let Parliament decide whether or not we remain a member of the European Union	25729	Oct 2016
Protect grouse Moors and grouse shooting	25322	Oct 2016
Walk away now! We voted for a No Deal Brexit	23679	Jan 2019
Not to allow freedom of movement as part of any deal with the EU after Brexit	21537	Oct 2016
The 23 June should be designated as Independence Day, and celebrated annually.	21292	Oct 2016
Save British Steel making. Scunthorpe, Teesside, Port Talbot etc.	18429	Oct 2015
Abolish the TV Licence	15678	Nov 2017
No referendum on the final deal for the UK to remain in the European Union.	14438	Dec 2017
Introduce super visa category for parents of British citizens similar to Canada	14297	Jul 2018
PublicHoliday on Hindus special two religious occasions in a year:DiwaliandDussera	11854	Oct 2018
Stop the scathing cuts to the Police budget.	9947	Nov 2015
British citizens should be able to appeal the refusal of a family 'visit visa'	6730	Jul 2018
To have a second referendum on Britain leaving the EU	4173	Jan 2019
Stop the destructive 'building our future' office closure programme in HMRC	2585	Nov 2015
Stop The Genocide In Myanmar - UK Government Please Help!	1451	Apr 2018

S5. Topic control strategy validation

I validate the topic control strategy in three ways. First, if MPs speak predictably on certain issue areas, then models that include topics should have better predictive performance than models without this information. In figure S1 I show that including MP-specific topic effects clearly decreases the DIC of the model relative to a null model which omits these effects. Furthermore, the model with 45 topics performs best in terms of DIC, and I focus on the results from this model throughout the paper. Figure S2 presents the coefficients of interest from all 20 topic models. Regardless of the specific topic model, the results are statistically and substantively very similar.

Figure S1: DIC by topic



Second, table S7 lists the names of 6 MPs with the highest β_{ik} coefficients from 4 example topics. The results are encouraging as MPs are associated with topical areas on which we have strong intuitions that they should be active. For example, the listed MPs with the highest

Table S7: MPs with highest β_{ik} coefficients by topic

trade;eu;union;negoti;european	**nhs;hospit;doctor;patient;amp**
Liam Fox	Jeremy Hunt
David Davis	Ben Gummer
Greg Hands	Jon Ashworth
Robin Walker	Stephen Barclay
George Hollingbery	Philip Dunne
Hilary Benn	Heidi Alexander
syria;daesh;iraq;syrian;humanitarian	**pupil;school;teacher;academi;educ**
Michael Fallon	Nick Gibb
John Woodcock	Edward Timpson
Crispin Blunt	Angela Rayner
Hilary Benn	Justine Greening
Mike Gapes	Lucy Powell
Emily Thornberry	Anne Milton

coefficients for the "trade;eu;union;negoti;european" topic are all well-known campaigners on the issue of the relationship between the UK and the EU and include MPs from both the 'Remain' (Hillary Benn) and 'Leave' (Liam Fox, David Davis) camps. Similarly, the MPs appearing in the "nhs;hospit;doctor;patient;amp" topic also make sense in substantive terms: Jeremy Hunt, Stephen Barclay and Ben Gummer were the Secretary, Minister and Undersecretary of State for Health, respectively, during the time period, and Heidi Alexander and Jon Ashworth have both been the shadow Secretary of State for Health during this period.

Similar patterns hold for the MPs listed under the "syria;daesh;iraq;syrian;humanitarian" topic. During the study period, Michael Fallon was the Secretary of State for Defence, and Hilary Benn and Emily Thornberry have both been the shadow Secretary. Similarly, MPs who feature on the "pupil;school;teacher;academi;educ" topic held government roles in the Department for Education (Nick Gibb, Edward Timpson) or were members of the opposition front-bench focussed on education (Lucy Powell; Angela Rayner). Overall, these tables reveal that the text-based control strategy I employ has reassuring face validity: the procedure automatically associates MPs with topical areas on which we have strong intuitions to believe they should be active.

Finally, as described in the main text of the paper, the control strategy relies on how well we can approximate the topics of petition debates in the full sample of non-petition debates. To investigate this, table S8 shows, for each of the petition debates in the data, the titles of the three closest matching *non-petition* debates according to the cosine-similarity between the topic vectors. I calculate the cosine similarity between the topic vectors of each petition debate and every other non-petition debate in the data. I then rank the debates for each petition debate according to this metric and select the top three to display in table S8. Again, the results are reassuring as across almost all petition debates in the sample, the topic model clearly clusters petition debates with non-petition debates that focus on substantively very similar policy areas.

Linear regression estimates (table S9) show that a five-unit increase in the signature rate is associated with an increase of 12 percentage points in participation probability of petition debates. However, a five-unit increase in the signature rate is also associated with a 7 per-

centage point increase in participation in these closest matching *non-petition* debates. This reinforces the importance of including MP-specific topic effects: MPs are likely to be active on topics that are of greater interest to their constituents, even in the absence of the information provided by an e-petition.

Petition Debate	**Closest Matching Debates**
Refugee Crisis in Europe	Humanitarian Crisis in the Mediterranean and Europe; Refugee Situation in the Mediterranean; Humanitarian Aid: Refugees in Greece and the Balkans
NHS	BMA; Barking, Havering and Redbridge University Hospitals NHS Trust; North Middlesex University Hospital NHS Trust
Sgt Alexander Blackman	Iraq Historic Allegations Team; Armed Forces Veterans: Historic Allegations; Veterans and Soldiers: Statute of Limitations
Cannabis	Sunbed Use: Health Implications; Drugs Policy; Drug Consumption Rooms
Immigration	Immigration; Immigration and Social Security Co-Ordination Bill; Immigration
Term-time Leave	School Penalty Fines and Authorised Absence; Relationships and Sex Education; Term-Time Holidays
Junior Doctors' Contracts	Junior Doctors Contracts; Junior Doctors' Contract Negotiations; Junior Doctors Contracts
Steel Industry	UK Steel Industry; UK Steel Industry; UK Steel Industry
Policing	Police Grant Report; Police Numbers; Police Funding
HMRC Office Closures	HMRC: Building Our Future Plan; Royal Bank of Scotland; HMRC Closures
Sugary Drinks Tax	Childhood Obesity Strategy; Tobacco Control Plan; Reducing Health Inequality
ISIL in Syria	Syria; Syria; Syria
Neonicotinoids on Crops	UK Bee Population; National Pollinator Strategy; Driven Grouse Shooting
NHS Bursary	NHS Bursary; Routes into Nursing; NHS Bursaries
Donald Trump	Polish Anti-defamation Law; Britain in the World; Anti-Semitism
Student Maintenance Grants	Higher Education Funding; Education; University Tuition Fees
Small Businesses: Tax Reporting	Taxes on Small Businesses; Small Businesses; Making Tax Digital
Transitional State Pension Arrangements for Women	State Pension: Women born in the 1950s; State Pension Age Equalisation; State Pension Age for Women
Non-EU Citizens: Income Threshold	University Tuition Fees; Funding Higher Education; Student Loans Agreement
BMA	NHS; Barking, Havering and Redbridge University Hospitals NHS Trust; Junior Doctors' Contracts
Brain Tumours	Bowel Cancer Screening Age; Anti-TNF Drugs; Accelerated Access Review
Meningitis B Vaccine	Appropriate Me Treatment; Orkambi and Cystic Fibrosis; Phenylketonuria: Treatment and Support
EU Referendum Leaflet	Brexit Deal: Referendum; EU Referendum Rules; Leaving the EU: Parliamentary Vote
Fireworks	Cat Welfare; Dangerous Dogs Act: Staffordshire Bull Terriers; Exotic Pets Trade
Foreign Aid Expenditure	Venezuela: Political Situation; Central and East Africa; Yemen
EBacc: Expressive Arts Subjects	Instrumental Music Tuition; Children's Academic and Practical Skills; Improving Education Standards
School Penalty Fines and Authorised Absence	Term-time Leave; Relationships and Sex Education; Relationships and Sex Education
Student Loans Agreement	Higher Education Funding; Student Maintenance Grants; Funding Higher Education
EU Referendum Rules	Leaving the EU: Parliamentary Vote; Brexit Deal: Referendum; Leaving the European Union
Dog Meat	Domestic Ivory Market; Dog Meat Trade; Fireworks
UK Exit from the European Union	Leaving the EU; Article 50 of the Treaty on European Union; Leaving the EU: Residency Rights
Local Government Pension Scheme	Nuclear Decommissioning Industry: Pensions; British Airways; AEA Pension Scheme
Driven Grouse Shooting	Protection of Ancient Woodland and Trees; Coastal Flood Risk; Tree Planting
Police Dogs and Horses	Corrosive Substance Attacks; National Road Safety Targets; Service Animals
Free Childcare	Free Childcare; Free Childcare: Costs and Benefits; Nursery Sector: Sustainability
Child Cancer	Paediatric Cancers of the Central Nervous System; Specialist Neuromuscular Care and Treatments; World Cancer Day

Retail Store Closure: Boxing Day	High Street Retailers; Statutory Pubs Code and Pubs Code Adjudicator; Pubs Code and the Adjudicator
Non-recyclable and Non-compostable Packaging	Plastic-free Packaging; backbench business Fisheries Policy; Plastic Bottles and Coffee Cups
Agenda for Change: NHS Pay Restraint	NHS: Staffing Levels; Doctors and Nurses; NHS Staff Recruitment and Retention
Domestic Ivory Market	Cosmetics Testing on Animals; Dog Meat; Ivory Bill
President Trump: State Visit	Anglo-Polish Relations; Battle of Jutland Centenary; Donald Trump
Attacks on NHS Staff	Acute and Community Health; Gosport Independent Panel; Southern Health NHS Foundation Trust
Workplace Dress Codes	Unpaid Work Trials; Employment and Workers' Rights Bill; Gender Pay Gap
Child Abuse Offences	Poppi Worthington; Forced Marriage; Parole Hearings: Victim Statements
Car Insurance: Young People	Vehicle Technology and Aviation Bill; Transmission of data relating to charge points; Energy Prices
Badger Culling	Badger Culling/Bovine TB; Badger Cull; Sunbed Use: Health Implications
TV Licence Fee	BBC; BBC Charter Review; White Paper on the BBC Charter
Public Sector Pay	Public Sector Pay Policy; Wages; Public Sector Pay Cap
Brexit Deal: Referendum	Leaving the European Union; Leaving the EU: Parliamentary Vote; EU Membership: Second Referendum
Pension Equality for Women	State Pension Age for Women; State Pension Age Equalisation; State Pension Age
Enslavement of Black Africans	Modern Slavery: Libya; Illegal Migration; Refugee Crises
Childcare Vouchers	Free Childcare; Free Childcare; Childcare Bill
Leaving the European Union	Leaving the EU; European Union Act; EU Reform
Fireworks	Fireworks: Public Sales; Fireworks; Cat Welfare
Leaving the EU: Live Farm Animal Exports	Foie Gras Imports; Fur Trade; Animal Welfare
British Sign Language: National Curriculum	Deaf Children's Services; Autism: Educational Outcomes; Autism Support
Orkambi and Cystic Fibrosis	Appropriate Me Treatment; Brain Tumours; ME: Treatment and Research
Myanmar: Rohingya Minority	Rohingya Crisis; Burma; Oman, Uae and Iran
Privatisation of NHS Services	Barking, Havering and Redbridge University Hospitals NHS Trust; NHS; BMA
Minors entering the UK: 1948 to 1971	Windrush; Windrush; Home Affairs and Justice
Grenfell Tower Inquiry	Grenfell Tower Fire Inquiry; Fire Prevention and Safety; Grenfell Tower
Sale of Puppies	Exotic Pets Trade; Sale of Puppies; Welfare of Young Dogs Bred for Sale
Fur Trade	Foie Gras Imports; Animal Welfare; Puppy Smuggling
Leaving the EU: Parliamentary Vote	Leaving the European Union; Brexit Deal: Referendum; EU Referendum Rules
House of Lords: Abolition	Points of Order; Points of Order; Hereditary Peers
Football: Safe Standing	Football: Safe Standing Areas; Rugby World Cup 2015; Football Grounds: Safe Standing Areas
Pet Theft	Sale of Puppies; Dangerous Dogs Act: Staffordshire Bull Terriers; Dog Fighting
Family Visitor Visas	Access Rights to Grandparents; Grandparents' Rights: Access to Grandchildren; Forced Adoption in the Uk
Dangerous Dogs Act: Staffordshire Bull Terriers	Exotic Pets Trade; Cat Welfare; Sale of Puppies
Vote Leave Campaign: Electoral Law	Leaving the EU: Rights of EU Citizens; EU Renegotiation; EU Exit: Article 50
Racehorse Protection	Dog Fighting; Dangerous Dogs Act: Staffordshire Bull Terriers; Cat Welfare
Autism and Learning Disability Training: Healthcare Professionals	Autism Diagnosis Waiting Times; - World Autism Awareness Weekbackbench Business; Transforming Care Programme
Public Holidays on Religious Occasions	International Day Against Homophobia, Transphobia and Biphobia; World Hijab Day; Women and the Vote
Plastic-free Packaging	backbench business Fisheries Policy; Recreational Sea Bass Fishing; Plastics: Agriculture
Leaving the European Union	Leaving the EU: Parliamentary Vote; Leaving the EU; Brexit Deal: Referendum
Fireworks: Public Sales	Fireworks; Fireworks; Cat Welfare
EU Membership: Second Referendum	Leaving the European Union; Brexit Deal: Referendum; Leaving the EU: Parliamentary Vote
Televised Election Debates	Electoral Commission; Advisory Committee on Business Appointments/Ministerial Code; Parliamentary Commissioner for Standards

Leaving the EU	Leaving the European Union; Leaving the EU: Parliamentary Vote; UK-EU Renegotiation
College Funding	Further Education; Technical and Further Education Bill; Apprenticeships
Cervical Cancer Smear Tests	Prostate Cancer; Surgical Mesh; HPV Vaccinations for MSM
Leaving the European Union	EU Membership: Second Referendum; Leaving the EU: Parliamentary Vote; Brexit Deal: Referendum
Secondary School Opening Hours	backbench business School Funding; Education Funding: Devon; English as an additional language
Relationships and Sex Education	Relationships and Sex Education; Statutory Sex and Relationships Education; School Penalty Fines and Authorised Absence

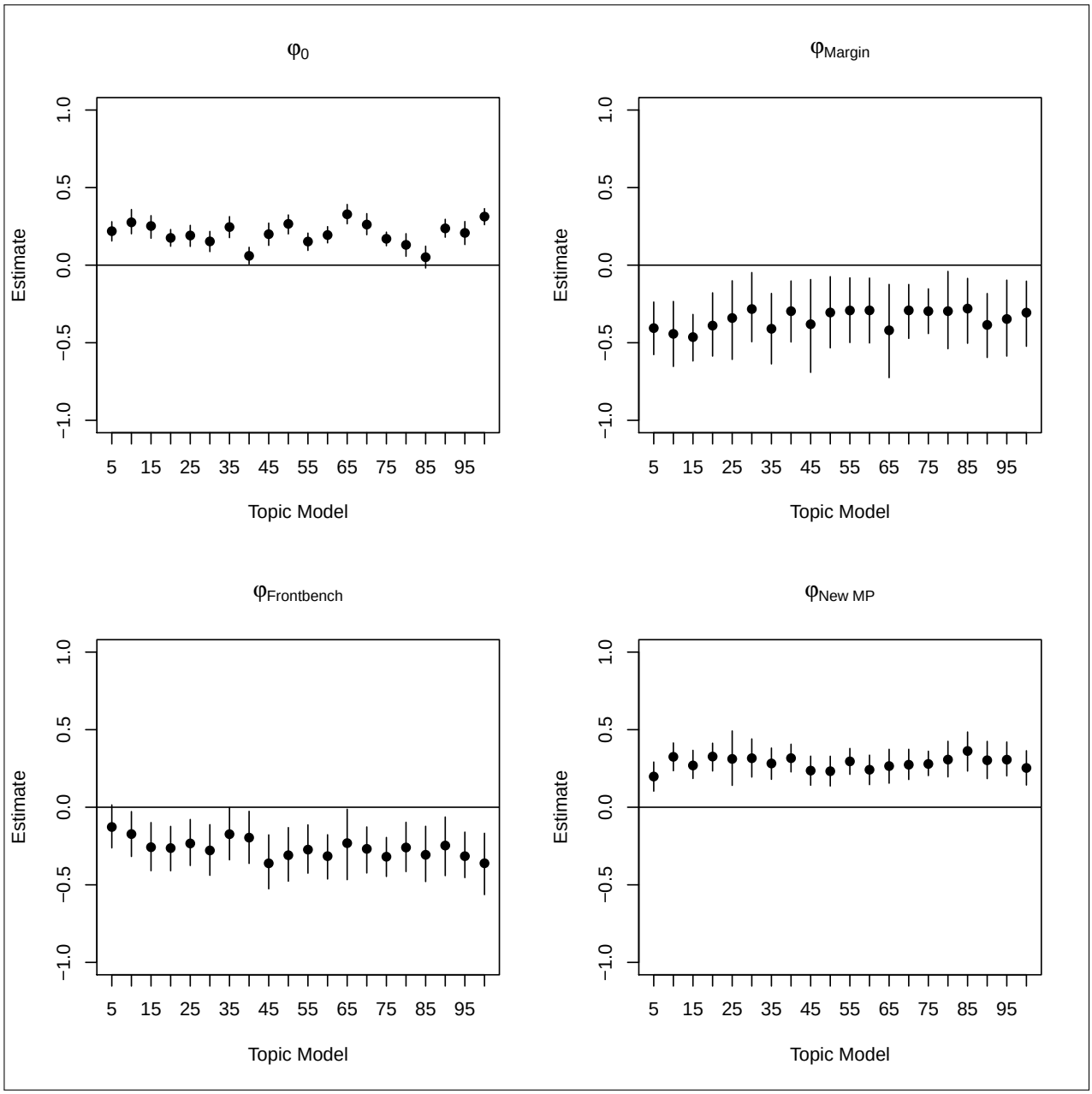
Table S8: Closest matching debates

Table S9

	Participation	
	Petition Debates	Closest matching non-petition debates
	(1)	(2)
Relative Signature Rate	0.025***	0.014***
	(0.001)	(0.002)
Constant	0.040***	0.080***
	(0.001)	(0.001)
Observations	52,733	52,733
R^2	0.009	0.001
Note:		*p<0.1; **p<0.05; ***p<0.01

Overall, the topic-control strategy improves our ability to predict MP debate participation; associates MPs to topics in ways that are consistent with intuitions about British politics; and links petition-debates to counterfactual non-petition-debates that are very similar in terms of their topical content.

Figure S2: Second-level predictors (debate participation), all topic models



Note: **Participation results are insensitive to topic-model specification**
The figure depicts the main second-level coefficients of interest (φ_0, $\varphi_{\text{Frontbench}}$, $\varphi_{\text{New MP}}$ and φ_{Margin}) from equation 3 for each of the topic models I estimate. The x-axis of each sub-panel gives the number of topics, and the y-axis gives the value of the relevant coefficient. Regardless of the number of topics estimated in the CTM model, the results are substantively and statistically very similar.

S6. Alternative independent variable specifications

To assess the degree to which the results presented in the body of the paper are sensitive to the specification of the main signature rate variable defined in equation 1 I present a series of bivariate logistic regression models below in table S10. Each model in the table presents the effect of petition signatures on debate participation, but where the exact measure for petition signatures varies across models. In particular, I test each of the following specifications for the signature rate:

$$X^1_{id} = \frac{\text{Constituency Signatures Per Capita}_{id}}{\text{National Signatures Per Capita}_d} - 1 \tag{S1}$$

$$X^2_{id} = log\left(\frac{\text{Constituency Signatures Per Capita}_{id}}{\text{National Signatures Per Capita}_d}\right) \tag{S2}$$

$$X^3_{id} = log\left(\text{Constituency Signatures}_{id}\right) \tag{S3}$$

$$X^4_{id} = \text{Constituency Signatures Per Capita}_{id} \tag{S4}$$

$$X^5_{id} = \frac{\text{Constituency Signatures Per Capita}_{id}}{\text{Average Constituency Signatures Per Capita}_i} - 1 \tag{S5}$$

Equation S1 is the measure I use in the paper, and is included here as a baseline comparison. Equation S2 is the log of the signature rate variable used in the paper. Equation S3 is the log of the number of signatures from a given constituency on a given petition. Equation S4 is the number of signatures from a given constituency on a given petition, divided by the number of electors in that constituency in 2015. Finally, equation S5 is the same as equation S1, but here rather than normalising the signature rate of a given constituency on a given petition by the national signature rate on that petition, I instead normalise by the average constituency rate across all petitions. Finally, for the purposes of comparison, I standardise each of these variables to have a mean of zero and a standard deviation of one.

As table S10 makes clear, regardless of how the signature rate variable is specified, I find a strong, positive, and significant relationship between constituency petition signatures and the probability of debate participation of the relevant MP.

Table S10: Bivariate logit – Participation

	Participation				
	(1)	(2)	(3)	(4)	(5)
Signature Variable	0.371^{***}	0.401^{***}	0.207^{***}	0.115^{***}	0.129^{***}
	(0.023)	(0.025)	(0.023)	(0.013)	(0.015)
Constant	-3.210^{***}	-3.246^{***}	-3.198^{***}	-3.188^{***}	-3.189^{***}
	(0.023)	(0.024)	(0.023)	(0.022)	(0.022)
Observations	52,735	52,735	52,735	52,735	52,735
Akaike Inf. Crit.	17,435.510	17,425.350	17,620.610	17,642.060	17,643.940

Note: *p<0.1; **p<0.05; ***p<0.01

S7. MP promotion of e-petitions via Twitter

To investigate the possibility that MPs use social media to drive their constituents to sign e-petitions that address issues on which the MPs are already active, I collect data on the tweets sent by MPs during the 2015-2017 parliamentary term from the Twitter API (https://developer.twitter.com). Not all MPs have twitter accounts, but I was able to identify the accounts of 554 MPs (out of 650) during this term. I collected all tweets sent by each MP between May 2015 and May 2017, resulting in just over 1 million tweets. I begin by taking a naive approach which almost certainly over-estimates the propensity for MPs to tweet about the petitions system by searching the strings of the tweets for the word "petition". Although any tweet about the petition system is likely to include this word, so will many other tweets that are not about the petitions system (such as tweets about local petitions, or petitions run through private organisations such as Change.org). Even with this permissive search criteria, the numbers of MPs making use of Twitter as a platform for promoting government e-petitions is very small. Of the more than 1 million tweets in my sample, only 1868 include the word "petition". Furthermore, of the MPs in my sample, a little over half ever mention petitions.

For a more conservative accounting of MPs' promotion of petitions from the parliamentary petition system, I search the strings of the tweets for links to the official e-petition website.[27] Again, while some MPs appear to use Twitter to promote e-petitions, the numbers are very small, as only 878 tweets include links to the parliamentary e-petition website. Further, of those MPs for whom I am able to collect Twitter data, only 258 ever include a link to the petitions website in their tweets during the study period. Given these small numbers, it seems highly unlikely that MP activism could explain the substantive results described in this paper.

[27] These are easily identifiable through the search string "`https://petition.parliament.uk/petitions/`"

Table S11: Second-level predictors (debate participation)

	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**
$\varphi_{\textbf{Margin}}$	-0.69 (-0.95, -0.45)	-0.39 (-0.63, -0.15)	-0.52 (-0.78, -0.23)	-0.48 (-0.69, -0.26)	-0.38 (-0.69, -0.09)
$\varphi_{\textbf{Frontbench}}$	-0.12 (-0.32, 0.1)	-0.15 (-0.42, 0.1)	-0.06 (-0.23, 0.11)	-0.11 (-0.28, 0.08)	-0.36 (-0.53, -0.18)
$\varphi_{\textbf{New MP}}$	0.29 (0.14, 0.42)	0.37 (0.24, 0.51)	0.29 (0.17, 0.39)	0.33 (0.25, 0.42)	0.24 (0.14, 0.33)
φ_0	0.14 (0.08, 0.2)	0.02 (-0.05, 0.08)	0.25 (0.19, 0.31)	0.31 (0.25, 0.38)	0.2 (0.13, 0.27)
$\mu_{\textbf{Margin}}$		-0.08 (-0.24, 0.07)		-0.1 (-0.24, 0.04)	-0.22 (-0.38, -0.08)
$\mu_{\textbf{Frontbench}}$		-0.14 (-0.26, -0.02)		-0.19 (-0.32, -0.06)	-0.52 (-0.63, -0.41)
$\mu_{\textbf{New MP}}$		0.29 (0.2, 0.37)		0.27 (0.16, 0.36)	0.49 (0.44, 0.54)
μ_0		-0.98 (-1.01, -0.94)		-1.91 (-1.96, -1.87)	0.57 (0.52, 0.61)
$\phi_{\textbf{Petition}}$			0.23 (0.09, 0.36)	0.23 (0.07, 0.38)	0.2 (0.02, 0.38)
ϕ_0			0.97 (0.95, 0.99)	-1.57 (-1.59, -1.54)	-1.45 (-1.47, -1.42)
φ Party dummies	✓	✓	✓	✓	✓
μ Party dummies	×	✓	×	✓	✓
MP random effects	×	✓	×	✓	✓
Debate random effects	×	×	✓	✓	✓
MP-topic random effects	×	×	×	×	✓
N parameters	1276	2549	6772	8045	40901
DIC	889976.4	841711	845104.3	793486.7	726863.5

Note: Table presents median estimates and associated 95% posterior intervals for the second-level logit coefficients described in equations 3 and 4 for the debate participation analysis. Model 1 is a baseline model, models 2 and 3 introduce MP and debate random-effects respectively, and model 4 includes both sets of random effects. Model 5 includes the matrix of MP-topic random-effects (β_{ik}).

Table S12: Second-level predictors (petition approval)

	Model 1	**Model 2**	**Model 3**	**Model 4**
$\varphi_{\textbf{Margin}}$	0.29 (-0.27, 0.8)	0.41 (-0.21, 1.04)	0.82 (-0.03, 1.61)	0.19 (-0.52, 0.92)
$\varphi_{\textbf{Frontbench}}$	-0.05 (-0.78, 0.79)	-0.37 (-1.15, 0.4)	0.12 (-0.86, 1.14)	0.11 (-1.13, 1.26)
$\varphi_{\textbf{New MP}}$	0.05 (-0.17, 0.27)	0.09 (-0.1, 0.29)	0.45 (0.21, 0.71)	0.55 (0.25, 0.85)
φ_0	0.95 (0.75, 1.14)	1.02 (0.83, 1.21)	0.58 (0.35, 0.8)	1.03 (0.83, 1.25)
$\mu_{\textbf{Margin}}$		-1.04 (-1.36, -0.71)		-1.42 (-1.86, -0.96)
$\mu_{\textbf{Frontbench}}$		-0.31 (-0.77, 0.18)		-0.31 (-1.01, 0.39)
$\mu_{\textbf{New MP}}$		0.5 (0.36, 0.66)		0.35 (0.18, 0.52)
μ_0		-1.43 (-1.55, -1.31)		-1.68 (-1.81, -1.55)
φ Party dummies	✓	✓	✓	✓
μ Party dummies	×	✓	×	✓
MP random effects	×	✓	×	✓
Debate random effects	×	×	✓	✓
N parameters	796	1589	882	1675
DIC	4046.1	3636.3	2672.4	2698.3

Note: Table presents median estimates and associated 95% posterior intervals for the second-level logit coefficients described in equations 3 and 4 for the petition approval analysis. Model 1 is a baseline model, models 2 and 3 introduce MP and debate random-effects respectively, and model 4 includes both sets of random effects.

Variable	print
$\varphi_{\textbf{Margin}}$	0.23 (0.12, 0.33)
$\varphi_{\textbf{Frontbench}}$	-0.11 (-0.19, -0.03)
$\varphi_{\textbf{New MP}}$	0.1 (0.03, 0.18)
φ_0	0.01 (-0.02, 0.05)
$\mu_{\textbf{Margin}}$	-0.19 (-0.3, -0.07)
$\mu_{\textbf{Frontbench}}$	-0.4 (-0.52, -0.29)
$\mu_{\textbf{New MP}}$	0.65 (0.57, 0.72)
μ_0	-0.88 (-0.93, -0.83)
$\phi_{\textbf{Petition}}$	0.23 (0.03, 0.43)
ϕ_0	-0.79 (-0.82, -0.76)
MP random effects	✓
Debate random effects	✓
MP-topic random effects	✓
N parameters	40896

Table S13: **Second-level φ effects (debate participation) – without party control**
The table presents median estimates and associated 95% posterior intervals for the second-level φ coefficients described in equation 3, but here excluding the φ_{Party} coefficients.